ML PRACTICAL 1

Bank Customer Churn Prediction using Artificial Neural Network (ANN)

Aim

To build an Artificial Neural Network model to predict whether a bank customer is likely to leave the bank using customer information.

Theory

Customer churn occurs when a customer stops using a company's services. In this practical, an Artificial Neural Network (ANN) is used to predict customer churn. The model learns patterns from past customer records and classifies each customer into one of two classes: 0 (not exited) or 1 (exited).

Dataset and preprocessing

The notebook loads **Churn_Modelling.csv**. The target column is **Exited**. The columns RowNumber, CustomerId and Surname are removed because they are identifiers or names, not useful model inputs. Geography and Gender are converted into numeric columns using one-hot encoding. The data is split into 80% training and 20% testing data, using stratification to preserve class proportions.

Feature scaling

StandardScaler standardizes the input features using the training data's mean and standard deviation. The scaler is fitted on the training set and then applied to both training and testing sets. This helps keep feature scales comparable and avoids using test-set statistics during training.

ANN model architecture

Layer	Configuration	Purpose
Input	All processed features	Receives customer data
Hidden layer 1	64 neurons, ReLU	Learns patterns
Dropout	0.3	Helps reduce overfitting
Hidden layer 2	32 neurons, ReLU	Learns further patterns
Dropout	0.2	Helps reduce overfitting
Output	1 neuron, Sigmoid	Predicts churn probability

Important concepts

ReLU: ReLU(z) = max(0, z). It is used in hidden layers. **Sigmoid:** $\sigma(z) = 1 / (1 + e^{-z})$. It gives an output between 0 and 1. **Binary cross-entropy:** Measures the error between actual binary labels and predicted probabilities. **Adam:** Optimizer used to update network weights during training.

Training the model

The model is compiled with the Adam optimizer, binary cross-entropy loss and accuracy metric. It is trained for up to 50 epochs with a batch size of 32. A validation split of 20% is used. EarlyStopping monitors validation loss; if it does not improve for five epochs, training stops and the best weights are restored.

Prediction and evaluation

The model predicts probabilities for the test data. A probability of 0.5 or above is classified as 1 (exited); otherwise it is classified as 0 (not exited). The notebook evaluates the model using:

- **Accuracy:** Overall proportion of correct predictions.
- **Confusion matrix:** Shows true positives, true negatives, false positives and false negatives.
- **Precision:** How many predicted positives are actually positive.
- **Recall:** How many actual positives are correctly identified.
- **F1-score:** Harmonic mean of precision and recall.

Result

An ANN model is built and trained to classify bank customers as likely to exit or remain. Its performance is assessed using accuracy, confusion matrix and classification report. Enter the actual metric values from your Jupyter Notebook execution; they are not present in the supplied code export.

Conclusion

This practical demonstrates the use of an Artificial Neural Network for binary classification. Data preprocessing, feature scaling, dropout and early stopping support the training process, while multiple evaluation metrics help assess the model's predictions.

Based on the uploaded Bank_Churn notebook. No execution metrics have been assumed.